

MIGA E&S; PORTAL 2.0

Complete System Documentation

Environmental & Social Risk Management System
Multilateral Investment Guarantee Agency | World Bank Group

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1. Executive Summary

The **MIGA E&S; Portal 2.0** is an enterprise-grade web application developed for the Multilateral Investment Guarantee Agency (MIGA), a member of the World Bank Group. The portal serves as the central platform for managing the complete lifecycle of guaranteed investment projects with a strong emphasis on Environmental and Social (E&S;) risk management, compliance monitoring, and due diligence.

MIGA's mandate is to promote foreign direct investment in developing countries by providing political risk insurance and credit enhancement. To ensure that guaranteed projects adhere to international environmental and social standards, MIGA E&S; Specialists perform rigorous evaluations against IFC Performance Standards (PS1–PS8). This portal digitalizes and streamlines that entire process — from the moment a project enters the pipeline through to portfolio monitoring, reporting, and eventual shelving.

Key System Highlights

Platform	Web Application (React.js frontend + ASP.NET Core REST API)
Environment	DEV — migaportalclouddev.worldbank.org
Organization	MIGA — Multilateral Investment Guarantee Agency, World Bank Group
Primary Purpose	E&S; Risk Assessment, Project Lifecycle Management, Compliance Monitoring
Users	E&S; Specialists, Managers, Administrators, Project Teams
API Standard	REST / Swagger UI (localhost:44379/api/esrd/swagger/index.html)
Key Standards	IFC Performance Standards PS1-PS8, GBV Risk Framework, ESRR Rating System
Architecture	CQRS with MediatR — decoupled Query/Command handlers

2. System Overview

2.1 About MIGA

The Multilateral Investment Guarantee Agency (MIGA) is a member of the World Bank Group that promotes foreign direct investment in developing countries. It provides political risk insurance and credit enhancement to investors and lenders to reduce risks associated with conflict, expropriation, currency restrictions, and breach of contract. By doing so, MIGA aims to boost economic growth and reduce poverty in developing nations.

For every project it guarantees, MIGA is required to assess and mitigate potential environmental and social risks. This requires a structured, systematic process involving trained E&S; Specialists who evaluate projects against internationally recognized IFC Performance Standards.

2.2 Purpose of the Portal

The MIGA E&S; Portal 2.0 was built to replace and modernize legacy processes. It provides:

- A centralized digital workspace for all E&S; Specialists and managers
- Structured workflows for due diligence, risk scoring, and compliance tracking
- A complete audit trail of all project decisions and documentation
- Real-time dashboards and analytics for portfolio health monitoring
- Integration with backend systems via a robust REST API
- Role-based access control ensuring data security and appropriate permissions

2.3 Application Modules at a Glance

Module	Purpose
Dashboard	Central analytics hub; pipeline vs portfolio stats, ESRR distributions, AMR summaries
Projects in Pipeline	Manage unapproved projects undergoing due diligence and risk rating
Projects in Portfolio	Monitor active MIGA-guaranteed projects; track monitoring activities
Shelved Projects	View and manage terminated or withdrawn projects
Other Initiatives	Track non-project activities, WBG cross-collaboration, CAO policies
Reports	Generate High Risk, ESAP, AMR, monitoring, and Nature Finance reports

3. Dashboard Module

3.1 Overview & Navigation

The E&S Dashboard is the first screen E&S Specialists and managers see upon login. It presents a consolidated view of all project activities, risk distributions, and compliance metrics across the entire MIGA portfolio and pipeline.

■ **NOTE:** The dashboard can be filtered by member (individual specialist or all members). Projection data is marked with an asterisk (*) indicating estimates based on two months ahead of today.

Navigation Button	Function
ES Insights Dashboard	Opens detailed pipeline and portfolio analytics with graphical views
GBV Dashboard	Opens the Gender-Based Violence risk assessment dashboard
Unassigned Projects (6)	Quick count and link to projects with no assigned E&S Specialist
Filter	Filter all dashboard data by member, region, sector, or fiscal year
Export	Export dashboard data to Excel/CSV for offline reporting

3.2 Business Stage Summary

The graphical section displays key operational metrics across three project states:

Project State	Metric	Value (as of reference)
Projects in Pipeline	Total	130 projects
Projects in Pipeline	Active / Slow Moving / Lead	75 / 51 / 4
Projects in Portfolio	Total	243 projects
Projects in Portfolio	Knowledge Gap / Projected Gap	6% / 14%
Projects in Portfolio	Upcoming Expiry	18 projects
Unassigned Projects	Total / Unassigned / Closed	95 / 94 / 1

3.3 Project Distribution Charts

Two interactive donut charts display:

- Project Distribution by Management Group — MAS (113), EEI (66), FINCAP (60), INF (23+25), TROF (13+3) comparing PRI vs NH vs TF product types
- Project Distribution by Sector — Financial (83), Infrastructure (156), Agribusiness (7), Manufacturing (23), Mining (4), Oil & Gas (1), Services (23), Tourism (15), Construction (3), Trade Finance (3)

3.4 ESRR Score Distributions

ESRR(D) — Pipeline (75 Projects)	ESRR(M) — Portfolio (243 Projects)
ESRR NA: 0 No Rating: 63 Excellent: 0	ESRR NA: 20 No Rating: 6 Excellent: 2
Satisfactory: 0 Partly Unsatisfactory: 11	Satisfactory: 178 Partly Unsatisfactory: 33
Unsatisfactory: 1	Unsatisfactory: 4

3.5 Project Category Distribution (318 Total Projects)

Category	Count	Category	Count
A	57	FI-1	10
A/B	4	FI-1/FI-2	1
B	146	FI-A(D)	—
C	14	TF	—

3.6 MSV (Monitoring Site Visit) Summary

MSV Status (243 Projects)	Planned MSV (54 Projects)	Track B Portfolio (46 Projects)
Not Recommended: 189	MSV Recommended & Planned: 51	Total IFC Track B: 45
Recommended & Completed: 4	MSV Recommended & Planned TBD: 3	Outstanding AMRs (FY): 1
Recommended Not Completed: 50	—	—

3.7 AMR (Annual Monitoring Report) Summary

Metric	FY Count	CY Count
Projects in Portfolio	243	—
Projects in Portfolio with AMR	213	—
Projects with AMR required (current cycle)	0	0
AMR for new projects signed	0	0
AMR received to date	FY26 (R): 20 (D): 0	CY26 (R): 1 (D): 0
AMR reviewed to date	1	1
Outstanding AMRs	7	7
Projects without AMR due date	0	0

4. Projects in Pipeline Module

4.1 Concept & Purpose

Projects in Pipeline are investment projects that have applied for a MIGA guarantee but have **NOT** yet been approved and issued a contract. They are in the evaluation phase. E&S; Specialists are assigned to each pipeline project to conduct Environmental & Social Due Diligence (ESDD) — a thorough assessment against IFC Performance Standards.

■ **NOTE:** Pipeline projects carry MIGOP statuses: PIPELINE ACTIVE, LEAD, or SLOW MOVING — indicating the commercial probability and deal velocity.

4.2 List View — Column Definitions

Column	Description
Project ID	Unique numeric identifier (e.g., 14383, 14840, 15046)
REG.#	MIGA Registration Number (e.g., 16030, 17298)
Charge Code	Internal budget code (e.g., MG-14383-SCDES-BB)
E&S DD Stage	Due diligence stage: ESDD Questionnaire, E&S clearance memo, Contract Negotiation, ESRS prepared, ESRS disclosed
Fiscal Year	MIGA fiscal year (2020-2027 range observed)
Region	World Bank region: Sub-Saharan Africa, Middle East & North Africa, South Asia, East Asia & Pacific, Latin America & Caribbean
Host Country	Country where the investment project is located
Guarantee Holder	Financial institution or organization holding the MIGA guarantee
Project Name	Name of the investment project
Lead E&S	Primary E&S Specialist assigned to the project

4.3 Sub-Tabs

My Projects

- Shows only projects assigned to the currently logged-in E&S; Specialist. Helps specialists manage their personal workload.

All Projects (91+ results)

- Comprehensive view of all active pipeline projects regardless of assignment. Useful for managers monitoring the full pipeline workload.

Unassigned Projects (16 shown)

- Shows pipeline projects with no assigned E&S; Specialist — representing unmanaged workload risk. Managers use this tab to assign projects. Examples include entries from Sub-Saharan Africa (Burundi, Burkina Faso, Congo), Europe & Central Asia (Kazakhstan), with guarantee holders such as Komatsu Financial Europe, Carbon Capture Singapore, China Energy International.

Master Contract Projects (1 shown)

- High-priority, fast-moving projects tracked separately. Example: Project 14840 (Raxio Data Centre) in Uganda, FY 2025, guaranteed by Meridiam Africa Investments, E&S; DD Stage: E&S; clearance memo.

Button/Feature	Function
Export	Download current filtered list as Excel/CSV for offline reporting
Search bar	Keyword search across project names, IDs, and charge codes
Columns	Toggle which columns are visible in the list view
Filters	Apply advanced filters by Region, Fiscal Year, E&S DD Stage, Guarantee Holder
Status tags	Color-coded: PIPELINE ACTIVE (teal) / LEAD (blue) / SLOW MOVING (orange)

5. Project Detail — Pipeline Stage

Clicking any project opens a multi-tab detail view. All tabs are described below using **Project 14383: ABSA/Barclays Botswana** as the reference example.

Project ID	14383
Project Name	ABSA/Barclays Botswana
Pipeline Status	Pipeline Active
Management Group	Finance and Capital Markets
Project Enterprise	ABSA Subsidiaries
Region	Sub-Saharan Africa
Host Country	Botswana
Sector / Sub-Sector	Financial / General Banking
Project Stage	Due Diligence
Underwriter	Calabozo Garrido, Olga
ESM Date	02/25/2019
PRC Date	05/09/2019
Charge Code	MG-14383-SCDES-BB
GBV Risk Rating	Medium FI
GBV Rating Status	Final

5.1 Tab: Project Info

Displays all general metadata including Project ID, Project Name, Management Group, Project Enterprise, Region, Host Country, Sector, Sub-Sector, PR/NH, SIP/Non-SIP, Knowledge Gap, ESM Date, PRC Date, Charge Code, Contract ID, Contract Effective/Expiry Date, Guarantee Holder, GBV Score and Risk Rating. Read-only for most users; Edit button available for authorized users.

5.2 Tab: Due Diligence

The heart of the E&S; assessment workflow. Captures the full Environmental and Social Risk Rating (ESRR) through a structured multi-section questionnaire.

5.2.1 Performance Standard List (PS1–PS8)

PS#	Performance Standard Name	Applies (Yes/No)
PS 1	Assessment and Management of E&S Risks and Impacts	Yes (primary)
PS 2	Labor and Working Conditions	Yes / No
PS 3	Resource Efficiency and Pollution Prevention	Yes / No
PS 4	Community Health, Safety and Security	Yes / No
PS 5	Land Acquisition and Involuntary Resettlement	Yes / No
PS 6	Biodiversity Conservation & Sustainable Management of Living Natural Resources	Yes / No
PS 7	Indigenous Peoples	Yes / No
PS 8	Cultural Heritage	Yes / No

- Additional questions: Does this project require alternative standards beyond IFC General/sector-specific EHS guidelines? (Yes/No)
- Are sector-specific EHS Guidelines applied? (Yes/No)
- Overall Comments: Free-text field for specialist narrative assessment

5.2.2 PS Achievement Rating Scoring

Specialists score Management, Performance, and Communication factors (each 1–3). Weighted combination generates the final ESRR(D).

Factor Group	# Questions	Areas Assessed
Management Factors	3	E&S Staffing, E&S Management System, Management Commitment
Performance Factors	5	Incidents/Compliance, PS/Safeguards Gaps, ESDD Quality, E&S Improvements Monitoring,
Communication Factors	2	Quality/Quantity of E&S Information, Stakeholder Communication

■ **NOTE:** Scoring: 1 = Poor, 2 = Average, 3 = Good. The 'Assess ESRR' button recalculates the final rating. An 'Override' button allows managers to manually override with justification.

5.3 Tab: ESAP (Environmental & Social Action Plan)

Manages the contractual action plan that the guarantee holder must fulfill as a condition of the MIGA guarantee.

Field	Description
Is ESAP Required?	Yes / No toggle — determines if an ESAP applies to this project
Action	Specific compliance action the contractor must complete
Due Date	Original target completion date
Re-Scheduled Date	Rescheduled date if the original was missed
Completion Date	Actual verified completion date

Field	Description
Deliverables	Expected output/evidence of completion
Remarks	Specialist notes and observations

- Buttons: Action Plan (download template), Template (blank), Upload (attach documents), Add (new action), Delete (with confirmation)

5.4 Tab: Documents

Manages all E&S-related documents for the project. The ESAP Documents section includes Document Type, Document Name, Status, and Document Notes. Example documents observed: Audit_Log-10-16-2025 (1).pdf, RiskDetails-Risk Inputs.xlsx, Aug14-SOE-Detail-Default.png, Agencia Financiera de Desarroll...

- Add ESAP Document button (top right), Delete button with checkbox selection

5.5 Tab: Nature Finance

Tracks the project's potential and actual contributions to nature-based financing frameworks, aligned with global biodiversity and climate commitments. Fields: Nature Finance (applicable or not), Nature Finance Potential (impact score), Justification (rationale for assessment).

5.6 Tab: GBV Risk Assessment

A specialized risk screening tab evaluating the project's risk of facilitating or failing to prevent Gender-Based Violence in the host country and project context. Follows a structured questionnaire with risk-rated responses, generating a final GBV Risk Score, GBV Risk Rating (e.g., Medium FI), and GBV Rating Status (Final/Pending).

5.7 Tab: Audit Summary

Read-only change log of all modifications made to the project record — who made the change, what was changed, and when — providing full traceability and compliance auditability.

6. Projects in Portfolio Module

6.1 Concept & Purpose

Projects in Portfolio are investment projects that have successfully received a MIGA guarantee and are now actively operational. Unlike pipeline projects (under evaluation), portfolio projects require **ongoing Environmental and Social monitoring** to ensure continued compliance with the agreed ESAP and IFC Performance Standards throughout the guarantee life.

■ **NOTE:** The portfolio contains 243+ active projects (250 results displayed as of reference). E&S; Specialists conduct annual AMR reviews, periodic site visits, and ongoing ESAP follow-up for each assigned project.

6.2 Portfolio List View

Project ID	Contract#	Region	Host Country	Project Name	Lead E&S
1258	11856-01	Sub-Saharan Africa	South Africa	HBZ Bank Limited	Abass, Emiola
3943	11955-01	Middle East & N. Africa	Jordan	As Samra	Diomampo, Catherine
4138	12177-01	Sub-Saharan Africa	Cote d'Ivoire	UT Abidjan	Sidore, Alessandro
5326	15771-01	Europe & Central Asia	Ukraine	Aluminum Beverage Can F...	Shrawne, Srujani
6732	14703-01	Sub-Saharan Africa	Uganda	Bujagali Hydroelectric Proj...	Zanewich, Debra
9286	10306-01	Sub-Saharan Africa	Ghana	Seawater Desalination Proj...	Ndong, Robert
9371	10401-01	Sub-Saharan Africa	Zambia	Hitachi Construction Mach...	Zanewich, Debra

7. Project Detail — Portfolio Stage

Portfolio project detail uses **Project 1258: HBZ Bank Limited** as the reference example.

Project ID	1258
Project Name	HBZ Bank Limited
Region	Sub-Saharan Africa
Host Country	South Africa
Sector	Financial
PR/NH	PRI
PRC Date	02/23/2012
Charge Code	MG-1258-SCCES-BB
Contract ID	11856-01
Contract Effective Date	03/30/2012
Contract Expiry Date	03/29/2022
Guarantee Holder	Habib Bank AG Zurich
GBV Risk Rating	Medium FI

7.1 Tab: Monitoring

The most critical tab for portfolio management. Records all E&S; monitoring activities performed during the guarantee life.

Action buttons: Add New Monitoring Activity | Delete | Edit | Export

Field	Type	Description
Fiscal Year	Integer	MIGA fiscal year (e.g., 2025)
AMR Review (checkbox)	Boolean	Whether an Annual Monitoring Report review was conducted
Reporting Period From/To	Date range	Period covered by the AMR (e.g., 1/1/2022 to 12/31/2023)
AMR Carried Out By	String	Who conducted the review (e.g., MIGA)
AMR Due Date	DateTime	Date the AMR was due (e.g., 05/01/2025)
AMR Received Date	DateTime	Date the AMR was actually received (e.g., 5/1/2025)
Site Visit (checkbox)	Boolean	Whether a physical site visit was conducted
Site Visit From/To	Date range	Date range of the site visit
Virtual Monitoring (checkbox)	Boolean	Whether virtual monitoring was used
Virtual Monitoring From/To	Date range	Date range for virtual monitoring activity

Field	Type	Description
Virtual Monitoring Carried Out By	String	Who conducted virtual monitoring
Client Meeting (checkbox)	Boolean	Whether a client meeting was held
Client Meeting Date	DateTime	Date of the client meeting
Other (checkbox)	Boolean	Other type of monitoring activity
Additional Description	Free text	e.g., 'AMRs for Jan to Dec 2022 and Jan to Dec 2023'

7.2 Tab: High Risk

Field	Description
Is this project CAO complaint?	Yes/No — whether a CAO (Compliance Advisor Ombudsman) complaint has been filed
Rationale for High Risk	Free-text explaining why the project was classified as high risk
Key E&S Issues	Summary of primary environmental and social concerns
Important Recent Developments	Material changes affecting the project including stakeholder attention

7.3 Risk Horizon & Associated Contracts

Risk Horizon: Forward-looking view of potential E&S; risks over the remaining guarantee life. Used for proactive intervention planning and monitoring activity scheduling.

Associated Contracts: Lists all contracts linked to the project including contract ID, effective date, expiry date, and status. Portfolio projects may have multiple contract amendments or host-country sub-contracts.

8. Shelved Projects Module

The Shelved Projects module contains projects no longer active in MIGA's pipeline or portfolio. These have been formally closed, terminated, or excluded from the guarantee program. The list shows **181 results** with the same column structure as the pipeline.

Status	Meaning	Example Project
DECLINED	MIGA decided not to issue the guarantee	—
WITHDRAWN	Investor or guarantee holder withdrew the application	
PROJECT NOT GOING FORWARD	The investment project was cancelled or abandoned	
DESELECTED BY MIGA	MIGA proactively removed the project from consideration	

Example shelved projects: Fayat SAS (Cameroon, 2025), Ghazanfar Neft & Gas DMCC (Afghanistan, 2025), FCC Aqualia S.A. (Georgia, 2025), Meridiam Infrastructure Africa Fund (Kenya, 2024), Dolma Fund Management (Nepal, 2024), Areni Holdings LLC (Armenia, 2024), FCA Investments LTD (Somalia, 2024).

■ **NOTE:** *Shelved projects retain all their documentation, due diligence records, and history for compliance and audit purposes. They are read-only and cannot be edited.*

9. Other Initiatives Module

Tracks non-project activities and strategic cross-organizational workstreams. These are not investment guarantee projects but are important activities requiring tracking, accountability, and deadline management.

FY	Status	Description	Initiative Type	Deadline
2023	ACTIVE	WBG Working Group – Nature Positive Finance and E&S Collaboration	WBG/IFC Cross Collaboration	12/31/2023
2023	ACTIVE	MFI Working Group on SEA/SH	WBG/IFC Cross Collaboration	12/31/2023
2022	ACTIVE	Project Level Grievance Mechanism work	CAO Policy – Action Items	04/30/2023
2022	ACTIVE	Solar supply chain related to legal rider	WBG/IFC Cross Collaboration	06/30/2023
2023	ACTIVE	FI Product Books	WBG/IFC Cross Collaboration	06/30/2023
2023	ACTIVE	IFC/MIGA Remedial Action Approach	WBG/IFC Cross Collaboration	12/31/2023
2023	ACTIVE	Improving systems and procedures, including E&S	CAO Policy – Action Items	06/30/2024
2023	ACTIVE	Strengthening and formalizing E&S case handling	CAO Policy – Action Items	06/30/2024
2026	ACTIVE	Targeted review of the CAO Policy	CAO Policy – Action Items	02/28/2026

- WBG/IFC Cross Collaboration: Joint workstreams with IFC on shared E&S; topics
- CAO Policy – Action Items: Tasks from CAO policy reviews and recommendations
- Add button allows authorized users to create new initiatives

10. Reports Module

The Reports module provides a dropdown menu of pre-built reports that can be generated on demand, aggregating data across the pipeline and portfolio for management, compliance, and stakeholder reporting.

Report Name	Description
High Risk Report	All projects currently classified as High Risk with CAO complaint status and key E&S issues
High Risk Removal Report	Documents projects removed from the High Risk classification
Recently Updated Projects Report	Projects with recent changes (last 30/60/90 days)
ESAP Status Report	Status of all ESAP action items across portfolio: overdue, on track, completed
ESRR Scores For Portfolio Report	Aggregates ESRR(M) scores for all portfolio projects for compliance reporting
AMR Submission Report	Tracks AMR submission rates, outstanding AMRs, and overdue submissions
Monitoring Site Visits – Trend Report	Trend analysis of MSV completion rates over time
Monitoring Activity Report	Comprehensive log of all monitoring activities across the portfolio
Snapshot – Projects in Pipeline	Point-in-time snapshot of all current pipeline projects and DD stage
Snapshot – Projects in Portfolio	Point-in-time snapshot of portfolio projects and monitoring status
Major Incidents Report	Documents major E&S incidents affecting MIGA-guaranteed projects
Nature Finance Report	Aggregates Nature Finance assessments across pipeline and portfolio

11. Complete System Workflow

11.1 End-to-End Project Lifecycle

Step	Stage	Activity
1	Project Creation	New project enters pipeline. Basic metadata recorded: Project ID, REG.#, Region, Country, Guarantee Holder, Sector
2	Assignment	E&S Specialist assigned to the project. Unassigned projects appear in the Unassigned Projects tab.
3	Due Diligence	Specialist conducts ESDD: assesses PS1-PS8 applicability, scores Management/Performance/Communication factors
4	ESRR Generation	System calculates ESRR(D) based on questionnaire responses. Specialist may override with justification.
5	ESAP Creation	If compliance actions required, specialist creates ESAP items with due dates and deliverables.
6	GBV Assessment	Specialist completes GBV Risk Assessment. GBV Risk Rating finalized.
7	Document Upload	Supporting documents (ESIAs, ESAP templates, audit logs) uploaded to the project.
8	Approval Decision	MIGA management reviews and approves/declines. Approved projects move to Portfolio.
9	Portfolio Monitoring	Portfolio project monitored via AMR review, site visits, virtual monitoring, client meetings.
10	ESAP Follow-up	Specialist verifies contractor completion of ESAP actions during monitoring activities.
11	High Risk Review	If major issues arise, project flagged as High Risk. Additional oversight and reporting.
12	Contract Expiry/Shelving	When guarantee expires or project is terminated/withdrawn, it moves to Shelved Projects.

11.2 Portfolio Projects — Complete Flow (Sequence Diagram)

The application implements a clean MediatR-based CQRS architecture. The sequence diagram for 'Portfolio Projects – Complete Flow' describes these interactions:

Participant	Role in the Flow
UI (Portfolio Grid)	React.js front-end grid component the user interacts with
ProjectsController	ASP.NET Core API controller receiving HTTP requests from the UI
MediatR	Message bus decoupling the controller from handlers using Command/Query pattern
PortfolioProjectsQueryHandler	Query handler orchestrating business logic for fetching portfolio projects
PortfolioProjectsService	Business service layer calling multiple data sources
ProjectsRepository	Database access layer: retrieves Portfolio Projections (AMR, ESRR, Risk) and Project & Application Details
Integration Service	External integration: retrieves ES Score & Phase Lookups from integrated systems
CommonService	Shared service for system-wide lookups and utilities

11.2.1 Sequence Flow Steps

- **Step 1:** User clicks 'Portfolio Projects' in the UI (Portfolio Grid)
- **Step 2:** UI sends Send(PortfolioProjectsQuery) to the ProjectsController
- **Step 3:** ProjectsController uses MediatR to Resolve Handler
- **Step 4:** MediatR routes to PortfolioProjectsQueryHandler.GetPortfolioProjects()
- **Step 5:** Handler calls PortfolioProjectsService
- **Step 6:** Service calls ProjectsRepository: Get Portfolio Projection (AMR, ESRR, Risk)
- **Step 7:** Service calls ProjectsRepository: Get Project & Application Details
- **Step 8:** Service calls Integration Service: Get ES Score & Phase Lookups
- **Step 9:** Service calls CommonService for additional lookups
- **Step 10:** Service executes Merge DB Projection + Integration (internal data merge loop)
- **Step 11:** Handler returns List to the controller
- **Step 12:** Controller sends PortfolioProjectsResponse back to the UI
- **Step 13:** UI renders the Show Portfolio Grid with the fully merged, enriched data

12. REST API Documentation

12.1 API Overview

The MIGA E&S Portal 2.0 exposes a comprehensive REST API documented via Swagger UI at **localhost:44379/api/esrd/swagger/index.html**. All responses follow a consistent envelope format:

Response Field	Type	Description
STATUS	String	Status message (e.g., 'Success', 'Error')
STATUS_CODE	Integer	Numeric status code (0 = success)
NOTIFICATIONS	Array	Notification objects with Code, Message, NotificationType (200 = info)
Result	Object/Array	The actual data payload — varies by endpoint

12.2 Projects API Group

Method	Endpoint	Description
POST	/api/esrd/Projects/GetPipelineProjects	Fetch pipeline projects — filters: FiscalYear[], ManagementGroupCode[], ProductType[]
POST	/api/esrd/Projects/GetPortfolioProjects	Fetch portfolio projects — same filter structure plus ESMFiscalYear[]
GET	/api/esrd/Projects/GetProjectDetails	Fetch single project by EsId (int), IsProjectsInPipeline (bool), IsShelvedProject (bool), or
GET	/api/esrd/Projects/GetShelvedProjects	Fetch all shelved projects via projectsQuery object parameter
POST	/api/esrd/Projects/GetGbvDetails	Fetch GBV assessment — body: ProjectId, EsId, HostCountryKey, IsProjectInPipeline
POST	/api/esrd/Projects/SaveGbvQuestionnaires	Save GBV responses — body: EsId, GbvId, ProjectNumber, GbvRiskRatingStatusCode
POST	/api/esrd/Projects/SaveGbvActionPlanDetail	Save GBV action plan — body: GbvActionPlanDetailId, GbvId, EsId, TypeOfAction, Exp
POST	/api/esrd/Projects/SaveTeamInformation	Save team — body: EsId, ProjectNumber, EsSpecialistUpI, SupportSpecialistUpIList[], R
POST	/api/esrd/Projects/CreateTeamAssignments	Create assignment — body: IsPipelineProject, ProjectNumber, IsESRD, ESRDCommen
POST	/api/esrd/Projects/SaveAdditionalInformationForPortfolio	Save additional portfolio metadata
POST	/api/esrd/Projects/SaveAdditionalInformationForPipeline	Save additional pipeline metadata

12.3 Monitoring API Group

Method	Endpoint	Description
POST	/api/esrd/Monitoring/SaveMonitoringActivity	Create/update monitoring — multipart form with FiscalYear, AMRReviewFlag, AMR
POST	/api/esrd/Monitoring/SaveMonitoringActivityAndAmrHistory	Save monitoring activity along with AMR history record
POST	/api/esrd/Monitoring/SaveMonitoring	Save monitoring data (simplified endpoint)
GET	/api/esrd/Monitoring/GetMonitoringList	Get all monitoring activities for a project by EsId (int)

Method	Endpoint	Description
GET	/api/esrd/Monitoring/GetMonitoringDetails	Get specific monitoring record details by EsId and ProjectStageId
GET	/api/esrd/Monitoring/GetDueDateForProject	Get AMR due date for a project by EsId
POST	/api/esrd/Monitoring/DeleteMonitoring	Delete a monitoring activity record
POST	/api/esrd/Monitoring/DownloadMonitoringReport	Download a monitoring report as file
POST	/api/esrd/Monitoring/DownloadMonitoringListReport	Download bulk monitoring list report

12.4 EsapAction API Group

Method	Endpoint	Description
POST	/api/esrd/EsapAction/SaveEsapAction	Save ESAP action — body: EsapActionId, EsId, Action, DueDate, RescheduleDate, Comments
GET	/api/esrd/EsapAction/GetEsapActions	Get all ESAP actions for a project by EsId
GET	/api/esrd/EsapAction/GetEsapDetails	Get full ESAP including EsapRequiredInfo and EsapActions[]
GET	/api/esrd/EsapAction/GetEsapRequired	Check if ESAP is required for project by EsId
POST	/api/esrd/EsapAction/SaveEsapRequired	Save ESAP required flag — body: EsId (int), IsEsapRequired (bool)
POST	/api/esrd/EsapAction/DeleteEsapAction	Delete ESAP actions — body: esapActionIdList[]
POST	/api/esrd/EsapAction/UploadEsapAction	Upload ESAP file — multipart: EsId (int), UploadFile (binary)

12.5 DueDiligence API Group

Method	Endpoint	Description
GET	/api/esrd/DueDiligence/GetDueDiligenceDetails	Get full DD questionnaire data for EsId. Returns DueDiligenceQuestionnaire with Responses
POST	/api/esrd/DueDiligence/SaveDueDiligence	Save/update DD — large body with EsId, ProjectStageId, GeneralInformation, Questionnaire
POST	/api/esrd/DueDiligence/DeleteDueDiligence	Delete a DD record by ProjectStageId
POST	/api/esrd/DueDiligence/DownloadDueDiligenceReport	Download DD report as PDF/Word — body includes all questionnaire sections: Context, General Information, Questionnaire

12.6 Dashboard & EsInsights API

Method	Endpoint	Description
POST	/api/esrd/Dashboard/GetPipelineProjectsForDashboardDetails	Get pipeline dashboard data — filter: FiscalYear[], ManagementGroupCode[], ProductType[]
POST	/api/esrd/Dashboard/GetPipelineProjectsForDashboardDetailsForSpecialist	Get pipeline dashboard data scoped to a specific specialist
POST	/api/esrd/Dashboard/GetPortfolioProjectsForDashboardDetails	Get portfolio dashboard data — same filter structure
POST	/api/esrd/Dashboard/GetPortfolioProjectsForDashboardDetailsForSpecialist	Get portfolio dashboard data scoped to specific specialist
POST	/api/esrd/EsInsights/StartMalenaUpload	Initiate Malena AI document analysis upload

Method	Endpoint	Description
POST	/api/esrd/EsInsights/GetMalenaResult	Get results of Malena AI analysis
GET	/api/esrd/EsInsights/GetEsInsightDocuments	Get list of documents available for ES Insights analysis

13. System Architecture

13.1 Technology Stack

Layer	Technology / Framework
Frontend UI	React.js — Single Page Application with component-based grid views
API Controller	ASP.NET Core — RESTful API controllers (e.g., ProjectsController)
Messaging Bus	MediatR — CQRS pattern for decoupled query/command handling
Business Layer	Service classes (e.g., PortfolioProjectsService) with domain business logic
Data Access Layer	Repository pattern (e.g., ProjectsRepository) for all database access
Integration Layer	Integration Service for external E&S score and phase data lookups
Common Services	Shared CommonService for system-wide lookups and utilities
API Documentation	Swagger UI — auto-generated from API controller annotations
Hosting Environment	World Bank Cloud (migaportalclouddev.worldbank.org) — DEV environment

13.2 CQRS with MediatR Pattern

The system follows the CQRS (Command Query Responsibility Segregation) pattern using MediatR:

- Queries (read operations like GetPortfolioProjects) are handled by dedicated Query Handlers that fetch and aggregate data without side effects
- Commands (write operations like SaveMonitoringActivity) are handled by Command Handlers that enforce business rules before persisting changes
- The API Controller acts as a thin routing layer dispatching requests to MediatR without any business logic
- This separation makes the codebase highly testable, maintainable, and independently scalable

13.3 Complete Data Flow

#	Layer	Action
1	UI (React.js)	User clicks a button, submits a form, or navigates to a tab
2	HTTP Request	UI fires an HTTP POST or GET to the ASP.NET Core API endpoint
3	Controller	Receives request and dispatches a MediatR message (query or command)
4	MediatR Bus	Resolves the correct handler for the message type
5	Handler	Executes the use case, calling service layer and repositories
6	Service + Repository	Business logic applied; database queries executed; external integrations called
7	Data Merge	Multiple sources (DB + Integration Service + CommonService) merged

#	Layer	Action
8	Response	Structured response (STATUS + NOTIFICATIONS + Result) returned
9	HTTP Response	Controller serializes response and sends JSON back to the UI
10	UI Render	React.js re-renders the grid/form with the received data

13.4 Security & Access Control

Role	Permissions
E&S Specialist	View/edit own assigned projects; conduct due diligence; upload documents; save monitoring activities; create ESAP action
Manager / Senior Specialist	View all projects (all members); override ESRR ratings; approve High Risk classifications; access all dashboard analytics
Administrator	Full system access; manage user assignments; configure system settings; export all data
Read-Only User	View project details, monitoring records, and reports without any editing capability

14. Key Business Glossary

Term	Definition
AMR	Annual Monitoring Report — submitted by guarantee holder annually detailing the project's E&S performance
CAO	Compliance Advisor Ombudsman — independent IFC/MIGA oversight function handling E&S performance complaints
CQRS	Command Query Responsibility Segregation — architecture pattern separating read and write operations
E&S	Environmental and Social — refers to environmental and social impacts and management of investment projects
ESAP	Environmental and Social Action Plan — contractual list of actions a guarantee holder must complete for E&S compliance
ESDD	Environmental and Social Due Diligence — assessment conducted by E&S Specialists for pipeline projects
ESMS	Environmental and Social Management System — systematic framework for managing E&S risks at the project level
ESRR	Environmental and Social Risk Rating — risk rating assigned (D = Due Diligence phase, M = Monitoring phase)
ESRR(D)	ESRR at Due Diligence stage — assigned during pipeline assessment via scoring questionnaire
ESRR(M)	ESRR at Monitoring stage — updated during ongoing portfolio monitoring activities
ESRS	Environmental and Social Review Summary — summary document of E&S findings for a project
FI	Financial Institution — project type involving banks, investment funds, or other financial intermediaries
GBV	Gender-Based Violence — specific risk category assessed for all MIGA projects via a dedicated questionnaire
IFC	International Finance Corporation — sister organization to MIGA within the World Bank Group
MediatR	Open-source .NET library implementing the Mediator pattern, used for CQRS messaging in this system
MIGA	Multilateral Investment Guarantee Agency — World Bank Group member providing political risk insurance
MSV	Monitoring Site Visit — physical visit to a project site for E&S compliance verification
PS	Performance Standard — IFC's eight performance standards (PS1-PS8) for E&S risk management
PRI	Political Risk Insurance — the core guarantee product MIGA offers to investors
SEA/SH	Sexual Exploitation and Abuse / Sexual Harassment — a subset of GBV risks tracked separately
SIP	Strategic Initiative Project — categorization for high-priority strategic pipeline projects
TF	Trade Finance — a type of MIGA guarantee product
WBG	World Bank Group — umbrella comprising IBRD, IDA, IFC, MIGA, and ICSID

15. Appendix

Appendix A — IFC Performance Standards Summary

PS#	Name	Key Requirements
PS 1	Assessment and Management of E&S Risks	EIS/IA impacts, stakeholder engagement, ESMS, monitoring program, grievance mechanism, public disclosure
PS 2	Labor and Working Conditions	Fair treatment, safe working conditions, workers' organizations, prohibition of child and forced labor
PS 3	Resource Efficiency and Pollution Prevention	GHG emissions, pollution controls, hazardous materials management, waste minimization
PS 4	Community Health, Safety and Security	Infrastructure safety, emergency preparedness, security forces code of conduct
PS 5	Land Acquisition and Involuntary Resettlement	Compensation at replacement cost, livelihood restoration, Resettlement Action Plans (RAPs)
PS 6	Biodiversity Conservation & Sustainable Management of Natural Resources	Original habitat preservation, biodiversity offsets, sustainable use of natural resources
PS 7	Indigenous Peoples	Free, Prior and Informed Consent (FPIC), cultural heritage protection, benefit sharing
PS 8	Cultural Heritage	Protection of tangible and intangible cultural heritage, chance finds procedures

Appendix B — ESRR Rating Scale

Rating	Code	Meaning
Excellent	E	Project consistently exceeds all E&S performance standards requirements
Satisfactory	S	Project fully meets all applicable E&S requirements
Partly Unsatisfactory	PU	Project partially meets requirements; corrective actions needed within timeframe
Unsatisfactory	U	Project significantly fails E&S requirements; escalation to management required
No Rating	NR	Rating not yet assigned — early stage project or insufficient data available
ESRR NA	NA	Rating not applicable to this specific project type or structure

Appendix C — API Quick Reference

Base URL (DEV): <https://localhost:44379/api/esrd/>

API Group	Key Endpoints
Projects	GetPipelineProjects GetPortfolioProjects GetProjectDetails GetShelvedProjects GetGbvDetails SaveGbvQuestionnaire
Monitoring	SaveMonitoringActivity GetMonitoringList GetMonitoringDetails GetDueDateForProject DeleteMonitoring DownloadMonitoringReport
EsapAction	SaveEsapAction GetEsapActions GetEsapDetails GetEsapRequired SaveEsapRequired DeleteEsapAction UploadEsapAction
DueDiligence	GetDueDiligenceDetails SaveDueDiligence DeleteDueDiligence DownloadDueDiligenceReport
Dashboard	GetPipelineProjectsForDashboardDetails GetPortfolioProjectsForDashboardDetails ...ForSpecialist variants

API Group	Key Endpoints
EsInsights	StartMalenaUpload GetMalenaResult GetEsInsightDocuments